

CONDENSED EDITION • FIVE THREE-HOUR CLASSES

Getting a number you can defend

Framing, data quality, and the evaluation hygiene that decides whether anything you do in the next four sessions produced a real result.

Session 1 of 5 • Data Preparation for Machine Learning and AI

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2 h lecture + 1 h paper clinic • long-form reference:

Chapter 0, §0.1–§0.3

Open by asking how the room currently splits data. Almost everyone says random 80/20. Hold that answer – it will look different by the end of block B.

What you should be able to do by the end of today

DIAGNOSE

Find the leak in someone else's pipeline

Name the split design a claim requires, and say why

Apply the eight-type leakage taxonomy to a published paper

Spot preprocessing fitted outside the fold

MEASURE

Put a number on data quality

Turn the six quality dimensions into computable, failing tests

Choose a drift statistic and state its caveat

Audit a corpus instead of trusting its README

DECIDE

Commit before you touch the data

Fix the task, the unit of analysis, the metric and the deployment constraint

Derive the split and the grain from that unit

Report both the naive and the honest number

DEFEND

Write it down so a reviewer can check

Fill in a model info sheet, per leakage type

State what your validation gate asserts, and what makes it fail

Carry these habits into sessions 2–4

These four verbs recur in every session and in the assessment rubric.

RUN SHEET

How the three hours run

BLOCK	CLOCK	WHAT HAPPENS
Lecture A	0:00 – 1:05	Framing and data quality — why preparation is now the experimental variable, and how to make quality computable
Break	1:05 – 1:20	Coffee. Use it: the second block is the harder one.
Lecture B	1:20 – 2:10	Splitting and leakage — split designs, cross-validation, the eight leakage types, and the optimism gap
Paper clinic	2:10 – 3:00	Find the leakage: each student brings one applied-ML paper from their own field.

DECK CONTROLS

E edit any line in place · N speaker notes · O slide grid · T timer · / search · P print a handout with notes underneath each slide.

ORIENTATION

Why this chapter comes first

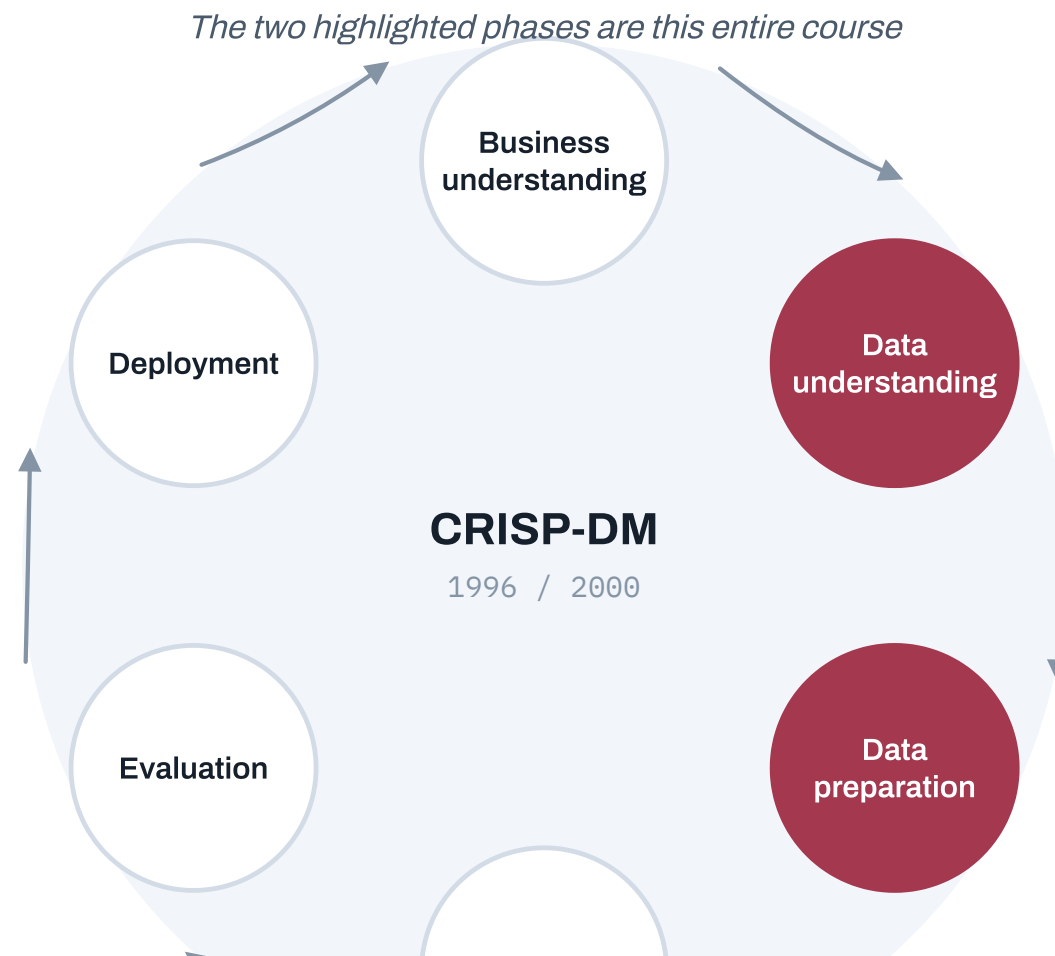
The tabular, text, and image chapters are technique catalogues. This one is the discipline that decides whether any of those techniques produced a real number.

Every later chapter has a step where a transform is fitted. **Where** you fit it is decided here.

Every later chapter has labels. **How much** you trust **them** is decided here.

Every later chapter reduces data — filtering, deduplication, pruning, resampling. Whether that reduction was a **sampling decision** with a **victim** is decided here.

And every later chapter



produces a
claim
someone

The two highlighted phases in the diagram are the whole course. Point out what CRISP-DM does NOT model: labels as an object of study, corpus-scale curation, and preparation as the main experimental variable.

CRISP-DM 1996/2000. Still the best shared vocabulary for where preparation sits. Its blind spots are the reason this course exists.

SESSION 1: A NUMBER YOU CAN DEFEND \$ 0.0 WHY THIS CHAPTER COMES FIRST

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0.1

Framing: where preparation sits

Before any technique: what problem are we actually solving, and against what metric?

The data-centric turn

Three-goal decomposition

CRISP-DM and its limits

The 80% folklore

Task-first design

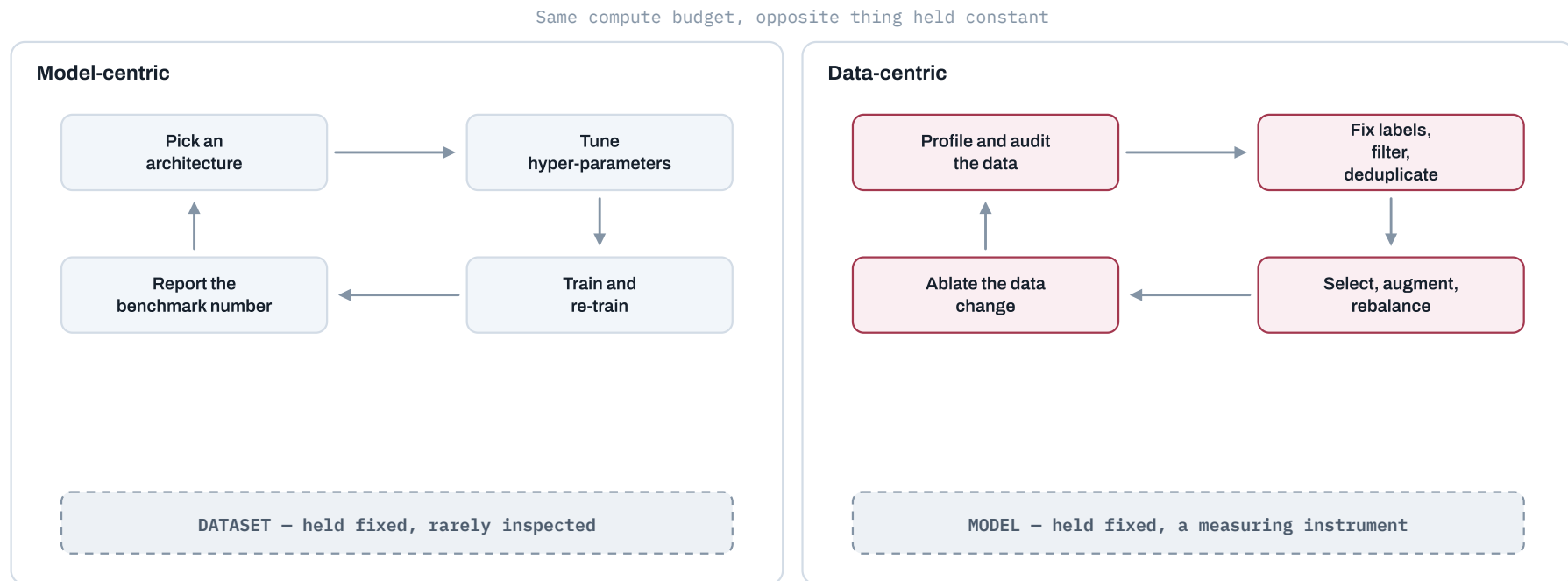
DC-Check

Automated preparation

0.1.1

The data-centric turn

Same compute budget. The difference is which of the two things you hold constant.



The right-hand loop is what this course teaches. Its evaluation discipline is stricter, because the thing you changed is the thing you measure with.

Andrew Ng's framing, formalised in the data-centric AI surveys. The claim is not that architecture stopped mattering — it is that at a fixed model, systematic data improvement is now the higher-yield lever, and it is far less explored.

Ask: in your last project, how many hours went to the left loop and how many to the right? Nearly everyone reports the left. That asymmetry is the opportunity.

Ablate (v.): to remove or destroy surface material through melting, vaporization, erosion, or surgery

Ablation Study (n.): กระบวนการทดลองในงานวิจัยด้านปัญญาประดิษฐ์ (AI) และการเรียนรู้ของเครื่อง (Machine Learning) โดยการถอดหรือปิดการทำงานของส่วนประกอบย่อยที่ละส่วนออก
SESSION 1 : A NUMBER YOU CAN DEFEND § 0.1 FRAMING: WHERE PREPARATION SITS
เปลี่ยนไป เพื่อดูว่าส่วนนั้นมีความสำคัญต่อประสิทธิภาพของโมเดลมากน้อยแค่ไหน

0.1.2

Three goals that get collapsed into one

Zha et al.¹ split data work into three problems with three different success criteria. Keeping them apart is most of the discipline.

Three engineering problems, three sets of metrics – do not let them collapse into one

Training-data development

- Collection and integration
- Labelling and label repair
- Cleaning, filtering, dedup
- Augmentation and synthesis
- Selection, pruning, mixing

Metric: downstream task performance at fixed compute

Inference-data development

- In-distribution evaluation sets
- Slice and subgroup sets
- Adversarial and stress sets
- Prompt / query sets for RAG
- Decontamination against training

Metric: does the eval set answer the question you are asking?

Data maintenance

- Quality monitoring in production
- Drift detection and alerting
- Versioning, lineage, rollback
- Storage, cost and access control
- Deletion and consent propagation

Metric: time-to-detect and time-to-repair

Most published “data-centric” gains are training-data work evaluated on an inference set nobody audited. That is the failure mode worth naming in week 1.

Teaching use: when a student says “I improved the data”, ask which of the three columns they mean, and which metric moved.

¹ Zha et al. *Data-centric Artificial Intelligence: A Survey*. ACM Computing Surveys, 2025.

The classic student failure: improve training data, evaluate on an inference set built from the same contaminated pool, and report a gain that overlap.

SESSION 1 . A NUMBER YOU CAN DEFEND S 0.1 FRAMING: WHERE PREPARATION SITS

FRONTIER

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The “80% of the time” claim, and how to replace it

The number is folklore repeated from industry surveys with small, self-selected samples. It is a useful conversation opener and a bad citation.

What it gets right: preparation dominates the calendar in most applied projects, and it is undervalued in publication.

What it hides: the fraction varies enormously by data family. A curated tabular benchmark is not a hospital PACS export.

Better move: measure it in your own project. Log hours against CRISP-DM phases for one term and report your own number in the capstone.

Even better: log *rework* hours separately — time spent redoing preparation after a downstream discovery. That is the number that motivates validation gates.

This is also the honest answer when a reviewer asks why a data paper deserves a slot: because the effort is real and currently invisible.

HABIT TO BUILD

Whenever you meet a striking quantitative claim in this course, ask three questions: who measured it, on what sample, and were they the same people who benefit from the answer? Several claims in the later chapters are the dataset authors' own ablations.

Task-first design, and DC-Check as a project gate

Every transform should be traceable to a decision you already made about the task. If it is not, you are decorating.

DECIDE FIRST

Four commitments before you touch the data

Downstream task — classification, ranking, segmentation, generation?

Unit of analysis — patient, visit, image, camera-hour, document, chunk?

Evaluation metric — and the decision it stands in for

Deployment constraint — latency, device, who acts on the output

THEN DERIVE

Each one constrains the preparation

The unit of analysis fixes the **split design** and the join grain

The metric fixes whether **calibration** matters or only ranking

The deployment constraint fixes the **preprocessing budget** at inference

The task fixes what counts as an **outlier** versus a rare event

THEN CHECK

DC-Check, stage by stage

A data-centric checklist answered *before* building, not after

Forces the uncomfortable questions early: provenance, subgroup coverage, drift plan

Works well as the first deliverable of a research project

Seedat¹, Imrie & van der Schaar, 2022

¹ Seedat & Imrie. *DC-Check: A Data-Centric AI Checklist to Guide the Development of Reliable Machine Learning...* 2022.

Automated data preparation, and why it lags AutoML

Model search is largely solved as an engineering problem. Preparation search is not, and the reasons are instructive.

The search space is not closed. A model space is a set of known architectures; a preparation space includes “construct a feature nobody has named yet”.

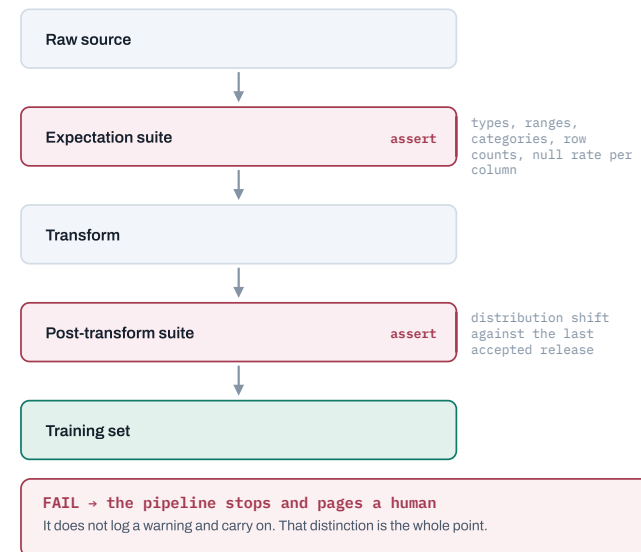
The objective is not scalar. Fidelity, fairness, privacy and cost all move, and a single validation score hides three of them.

Order matters and is not commutative. Impute-then-scale differs from scale-then-impute; filter ordering changes corpus composition.

Validation is fragile. The thing being searched is the thing that leaks — so a naive AutoML loop over preprocessing overfits the validation set faster than model search does.

Where it works today: constrained, well-typed pipelines with strong fold discipline. Treat published pipeline-search results with the same scepticism you apply to any tuned number.

A validation gate is a contract, not a report



Any automated search is only as honest as the gate it must pass through.

Section 0.1 — references

- 1 Zha, Bhat, Lai, Yang, Jiang, Zhong & Hu. **Data-centric Artificial Intelligence: A Survey**. *ACM Computing Surveys*, 2025 (arXiv:2303.10158).
- 2 Jakubik, Vössing, Köhl, Walk & Satzger. **Data-centric Artificial Intelligence**. 2024 (arXiv:2212.11854).
- 3 Seedat, Imrie & van der Schaar. **DC-Check: A Data-Centric AI Checklist to Guide the Development of Reliable Machine Learning Systems**. 2022.
- 4 Chapman et al. **CRISP-DM 1.0 Step-by-step data mining guide**. 2000; Fayyad, Piatetsky-Shapiro & Smyth, **From Data Mining to Knowledge Discovery**. *AI Magazine*, 1996.
- 5 Whang, Roh, Song & Lee. **Data Collection and Quality Challenges in Deep Learning: A Data-Centric AI Perspective**. *VLDB Journal*, 2023.
- 6 Mazumder et al. **DataPerf: Benchmarks for Data-Centric AI Development**. *NeurIPS Datasets & Benchmarks*, 2023.

0.2

Quality, profiling and validation

A quality dimension you cannot compute is an adjective. This section turns six adjectives into failing tests.

Six dimensions

Profiling

Schema contracts

Unit tests for data

Drift taxonomy

Drift statistics

Corpus auditing

0.2.1

Six quality dimensions, operationalised

Wang & Strong¹'s dimensions are forty years old and still the right list. The teaching move is to insist on the third line of each box.

A dimension you cannot compute is an adjective, not a quality dimension

Completeness

is anything missing that should be here?

MEASURE IT AS

count of nulls per column, per subgroup

Validity

does every value obey its declared type and range?

MEASURE IT AS

schema + constraint pass rate

Accuracy

does the value match reality?

MEASURE IT AS

audit against a reference standard

Consistency

do related fields agree with each other?

MEASURE IT AS

cross-field rule violations

Uniqueness

is each real-world entity represented once?

MEASURE IT AS

duplicate / linkage rate

Timeliness

is the value current enough to act on?

MEASURE IT AS

age distribution at prediction time

Adapted from Wang & Strong (1996) and Batini & Scannapieco. The right-hand test column is the part students skip.

¹ Wang & Strong. *Beyond Accuracy: What Data Quality Means to Data Consumers*. JMIS, 1996.

Completeness - การเติม Missing

Cut --> Dummy (999, -999) --> Mean/Mode/Median Imputation --> Data understanding --> Model to predict the missing value.

Validity - ความถูกต้องของ Data ตาม Metadata

Accuracy - ความแม่นยำ ความเหมาะสม เช่น น้ำหนักของ นศ. 1,000 กิโล

Consistency - ความสม่ำเสมอ หรือความไม่ซ้ำซ้อนของข้อมูล

Uniqueness - ความซ้ำของข้อมูลในแนว row เช่น ID ซ้ำกัน แต่ตัวแปรอื่น ๆ ไม่เหมือนกันหมดเลย (แล้วแต่ Dataset)

SESSION 1 · A NUMBER YOU CAN DEFEND § 0.2 DATA QUALITY, PROFILING AND VALIDATION

Timeliness - ข้อมูลนี้ยังเหมาะสมในการใช้ หรือไม่เช่น อายุ ผ่านไปตามกาลเวลา

TEXTBOOK

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Profiling: the first pass you never skip

Per column: type, cardinality, null rate, min/max, distribution shape, top values, and how many values are impossible.

Per pair: correlation, and the missingness co-occurrence map — columns that go missing together tell you about the collection process.

Per row: duplicate rate, near-duplicate rate, and completeness profile.

Per group: everything above, stratified by site, scanner, camera, clinician or time window. This is where site effects announce themselves.

Tooling: `ydata-profiling`, `Sweetviz`, `pandas.describe` for a first look; `Great Expectations`, `TFDV`, `Deequ` once you know what to assert.

The profiling report is *not* a deliverable. The assertions you write after reading it are.

TEACHING POINT

Tukey¹'s *Exploratory Data Analysis* (1977) argued for looking before modelling. The modern version of that argument is: look, then encode what you saw as a test that will fail next month when it stops being true.

¹ Tukey. *Exploratory Data Analysis*. Addison-Wesley. 1977.

Schema contracts and unit tests for data

Declare the contract: expected types, ranges, allowed categories, referential integrity, row-count bounds, and the acceptable null rate per column.

Assert twice: once on the raw source, once after transformation. A transform that silently produces NaNs passes the first gate and fails the second.

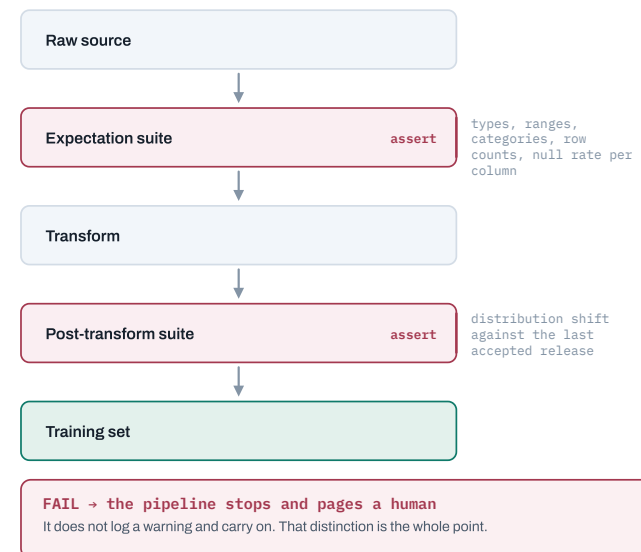
Fail loudly. A warning in a log is not a gate. The pipeline stops and a human is paged.

Treat data like code: assertions, fixtures, golden files, and CI that runs on every data refresh — the ML Test Score framing (Breck et al.¹, 2017).

Version the contract alongside the data. A relaxed threshold is a decision that should appear in a diff.

What to assert about distributions: not equality, but bounded divergence from the previous accepted release.

A validation gate is a contract, not a report



Two suites, not one: the raw gate catches upstream change, the post-transform gate catches your own bugs.

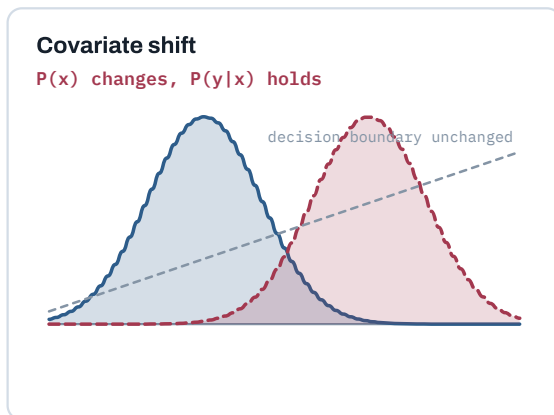
¹ Breck et al. *The ML Test Score*. IEEE Big Data, 2017.

0.2.5

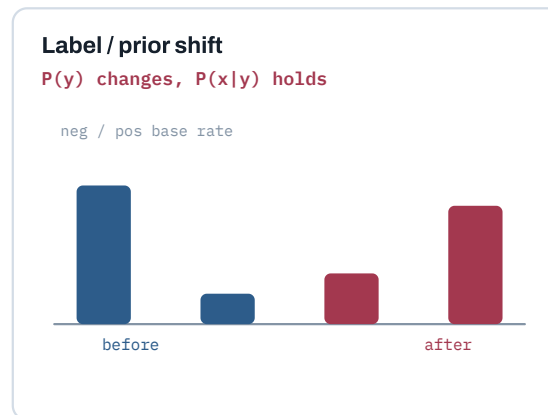
Three shifts, three responses

Covariate shift, prior shift and concept drift look identical on a dashboard. They demand different fixes.

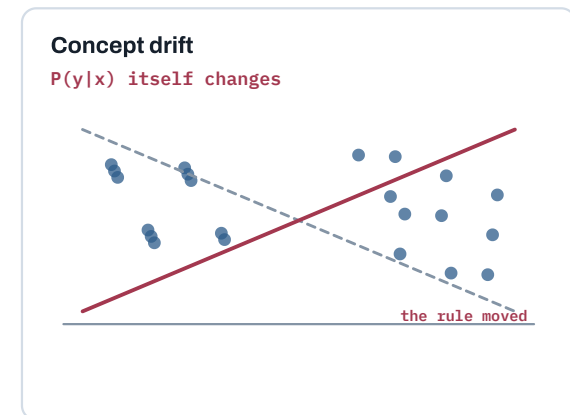
Three shifts that look identical on a monitoring dashboard and demand different responses



New scanner, new camera angle, new patient mix. The mapping is still valid — reweight or recalibrate.



Screening moves from a referral clinic to a village. Prevalence drops, thresholds must move.



Coding practice changes, a guideline is revised, an attacker adapts. Retraining is unavoidable.

The statistic that fires (PSI, KS, MMD, classifier two-sample) tells you something moved. It never tells you which of these three it was — that takes a hypothesis and a label.

Taxonomy after Quiñero-Candela et al.¹ (2009) and Moreno-Torres et al.² (2012). The distinction matters because only the third one forces you to relabel.

¹ Quiñero-Candela et al. *Dataset Shift in Machine Learning*. MIT Press. 2009.

² Moreno-Torres et al. *A unifying view on dataset shift*. Pattern Recognition, 2012.

Choosing a drift statistic — and its caveat

STATISTIC	WHAT IT COMPARES	WHERE IT MISLEADS
PSI (population stability index)	Binned marginal distributions	Bin choice dominates the value; industry thresholds (0.1 / 0.25) are conventions, not theory
Kolmogorov–Smirnov	Empirical CDFs, one dimension	Univariate only, and at $n > 10^5$ it flags differences far too small to matter
Chi-square	Categorical frequencies	Sensitive to rare categories and to sample size in the same way
MMD (kernel two-sample)	Full multivariate distributions	Kernel and bandwidth choice; expensive; hard to attribute to a column
Classifier two-sample test	Can a model tell old from new?	The most interpretable — the classifier's importances name the drifting feature — but it needs its own honest split

THE CAVEAT THAT MATTERS MOST

Every one of these is a hypothesis test. With enough rows, everything is significant. Monitor **effect size and downstream metric impact**, not p-values — and always ask whether the change is one you should adapt to or one you should reject at the gate.

Auditing what is actually in a corpus

For anything web-scale, the README is a hypothesis. Treat it as one.

Search it, do not trust it. Elazar et al.¹, *What's In My Big Data?* (ICLR 2024), built tooling precisely because nobody could answer basic questions about their own pretraining corpora.

Documented findings that should worry you: benchmark contamination, large volumes of machine-generated and duplicated text, and PII sitting in supposedly public data.

Dodge et al.² (2021) did the same for C4 and found blacklist filtering removing whole registers of legitimate text along with the noise.

Minimum audit for any corpus you use: sample 200 documents at random and read them. Yes, actually read them. It takes an afternoon and it changes every downstream decision.

Then quantify: language mix, source-domain mix, duplication rate, date distribution, and overlap with every benchmark you intend to report.

SEMINAR EXERCISE

Hand each student a different public dataset and one instruction: “find three things in it that its documentation does not mention.” Nobody has ever come back empty-handed.

¹ Elazar et al. *What's In My Big Data?*. ICLR, 2024.

This is the highest-value-per-minute exercise in the whole module.

Section 0.2 — references

- 1 Wang & Strong. **Beyond Accuracy: What Data Quality Means to Data Consumers.** *JMIS*, 1996.
- 2 Breck, Cai, Nielsen, Salib & Sculley. **The ML Test Score.** *IEEE Big Data*, 2017.
- 3 Quiñonero-Candela, Sugiyama, Schwaighofer & Lawrence. **Dataset Shift in Machine Learning.** MIT Press, 2009.
- 4 Moreno-Torres, Raeder, Alaiz-Rodríguez, Chawla & Herrera. **A unifying view on dataset shift.** *Pattern Recognition*, 2012.
- 5 Elazar et al. **What's In My Big Data?** *ICLR*, 2024.
- 6 Dodge et al. **Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus.** *EMNLP*, 2021.
- 7 Tukey. **Exploratory Data Analysis.** Addison-Wesley, 1977.
- 8 Tooling: Great Expectations, TensorFlow Data Validation, Amazon Deequ, ydata-profiling, Evidently, NannyML.

0.3

Splitting, leakage and evaluation hygiene

The anchor section of the whole module. If a student remembers one thing from this course, make it this.

Split designs

Cross-validation

Eight leakage types

Fitting inside folds

Duplicates

Temporal validity

Contamination

Model info sheets

0.3.1

The same data, split three ways

Only one of these designs answers the question a clinical or surveillance claim is actually making.

The same 24 rows, split three ways — only one of them answers the question you are asking

Random split

by image

A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H

■ train ■ test

Same patient appears in train and test. The model can memorise the patient, not the disease.

Grouped split

by patient

A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H

■ train ■ test

Every image of a patient lands on one side. This is the minimum for any clinical claim.

Temporal split

by acquisition date

A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H
A	B	C	D	E	F	G	H

■ train ■ test

Train on the past, test on the future. The only design that matches how the model will be used.

The tell: if held-out accuracy drops sharply when you move from the left panel to the middle one, the left panel was measuring patient identity.

Report both numbers in the paper. Reviewers who have read Kapoor and Narayanan will ask.

Grouping unit in our own work: patient for BiTNet ultrasound, test-kit specimen for OV-RDT, camera and session for CCTV.

Eight patients, three images each. The grouping unit is a modelling decision, not a technicality — and it must be declared before the first experiment.

- 1 Intharah et al. *BiTNet: Hybrid deep convolutional model for ultrasound image analysis of human biliary tract...* Artificial Intelligence in Medicine, 2023.
- 2 Kapoor & Narayanan. *Leakage and the Reproducibility Crisis in Machine-Learning-Based Science*. Patterns 4(9), 2023.

This is the slide to slow down on. Ask each group to name the grouping unit for their own project and write it on the board. Disagreements
SESSION 1 . A NUMBER YOU CAN DEFEND \$ 0.3 SPLITTING, LEAKAGE AND EVALUATION HYGIENE up are
the interesting part.

TEXTBOOK

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Cross-validation, and the variant you probably need

k-fold — the default. Fine when rows are exchangeable and plentiful, which in health data they almost never are.

Stratified k-fold — preserves class balance per fold. Mandatory under imbalance; still wrong if rows are grouped.

Group k-fold / leave-one-group-out — the correct default for patients, sites, scanners, cameras and sessions.

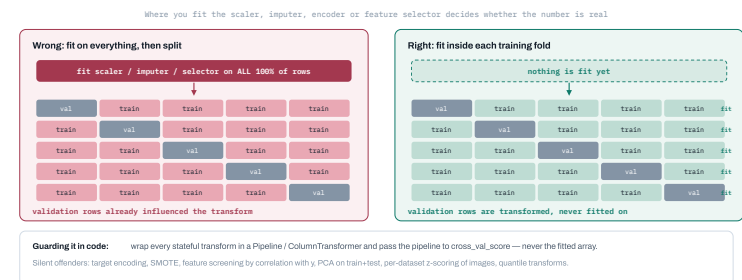
Nested CV — an inner loop for tuning, an outer loop for estimating. Without it, the reported score includes the model-selection search (Cawley & Talbot³, 2010).

Blocked / walk-forward — for anything with a time index. Covered in 0.3.8.

Repeated CV — reduces variance of the estimate; does *not* reduce bias from a wrong design. Repeating a leaky split just gives you a precise wrong number.

THE RULE OF THUMB

Pick the split that makes the training set *least* like the test set among all designs that are still honest. Optimism comes from similarity you did not intend.



Whichever variant you pick, the transform must be fitted inside the training portion of every fold.

¹ Chawla et al. *SMOTE: Synthetic Minority Over-sampling Technique*. JAIR, 2002.

- 2 Blagus & Lusa. *SMOTE for high-dimensional class-imbalanced data*. BMC Bioinformatics, 2013.
- 3 Cawley & Talbot. *On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation*. JMLR, 2010.

0.3.3

The eight-type leakage taxonomy

Kapoor & Narayanan¹ surveyed 22 reviews and found leakage in 17 fields, collectively affecting 294 papers.

#	TYPE	HOW IT SHOWS UP IN PRACTICE
L1.1	No test set at all	The reported number is training performance, sometimes without the authors realising
L1.2	Pre-processing on the full dataset	Scaler, imputer, encoder or PCA fitted before the split
L1.3	Feature selection on the full dataset	Screening by correlation with y across all rows, then cross-validating what survives
L1.4	Duplicates across the split	Exact or near-duplicate rows, images, frames or documents on both sides
L2	Illegitimate features	A feature that is a proxy for the target, or is only recorded after the outcome is known
L3.1	Temporal leakage	Training on data from after the prediction time; centred windows; future-looking aggregates
L3.2	Non-independence	Same patient, site, device, author or session on both sides of the split
L3.3	Sampling bias in test	The test set is not drawn from the population the claim is about

THE FINDING WORTH QUOTING

In their civil-war-prediction reproduction, once the leakage is corrected, complex ML models do **not** perform substantively better than decades-old logistic regression.

Print this table. It becomes the marking rubric for the paper-clinic seminar in Chapter 4.

0.3.5

Where you fit the transform

The single most common leak in applied machine learning, and the easiest to prevent in code.

Where you fit the scaler, imputer, encoder or feature selector decides whether the number is real

Wrong: fit on everything, then split

fit scaler / imputer / selector on ALL 100% of rows

val	train	train	train	train
train	val	train	train	train
train	train	val	train	train
train	train	train	val	train
train	train	train	train	val

validation rows already influenced the transform

Right: fit inside each training fold

nothing is fit yet

val	train	train	train	train	fit
train	val	train	train	train	fit
train	train	val	train	train	fit
train	train	train	val	train	fit
train	train	train	train	val	fit

validation rows are transformed, never fitted on

Guarding it in code: wrap every stateful transform in a Pipeline / ColumnTransformer and pass the pipeline to cross_val_score — never the fitted array.

Silent offenders: target encoding, SMOTE, feature screening by correlation with y, PCA on train+test, per-dataset z-scoring of images, quantile transforms.

The guard is architectural: wrap every stateful transform in a Pipeline / ColumnTransformer and hand the pipeline — never a pre-fitted array — to the cross-validator.

¹ Chawla et al. *SMOTE: Synthetic Minority Over-sampling Technique*. JAIR, 2002.

² Blagus & Lusa. *SMOTE for high-dimensional class-imbalanced data*. BMC Bioinformatics, 2013.

Ask them to name a stateful transform they have fitted outside a fold. Everyone has. The honest answer normalises the discussion and makes threatening.

SESSION 1 · A NUMBER YOU CAN DEFEND · S 0.3 · SPLITTING, LEAKAGE AND EVALUATION · HYGIENE

TEXTBOOK

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0.3.5

The optimism gap, measured

Leakage does not produce a small bias. It produces the difference between a publishable result and a real one.

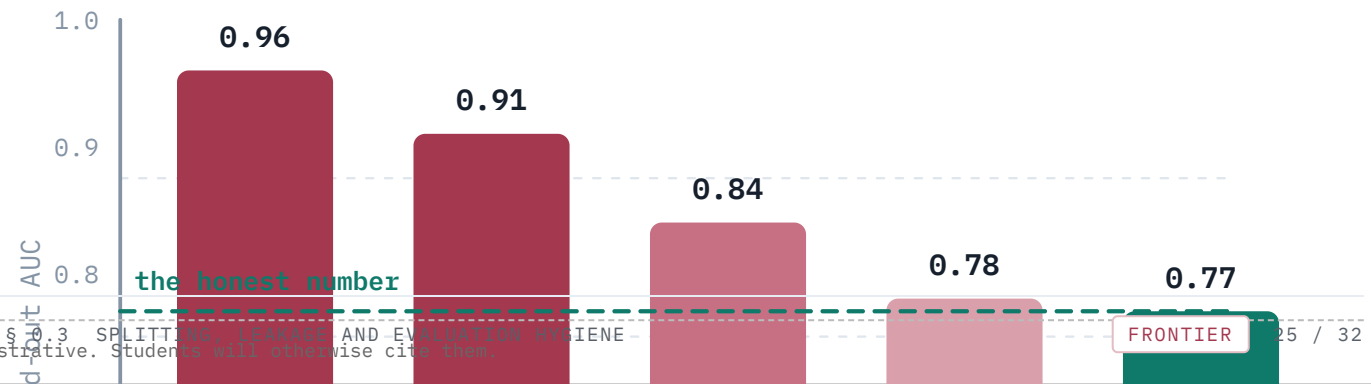
Every bar to the left of the dashed line is a **result someone has published**, in some field, in the last five years.

The gap is largest exactly where data is scarcest and grouping is strongest — which is to say, in clinical research.

Lab 1 is built on this **figure**: students fit the same model with the same data, but transform it inside

What leakage buys you

Illustrative shape of the optimism gap, not measured values — run it yourself in Lab 1.



0.3.6-0.3.7

Duplicates, and grouping in clinical and CCTV data

Near-duplicates are the silent accuracy inflator. Two frames one second apart, the same slide scanned twice, the same report copy-forwarded, the same web page under two URLs.

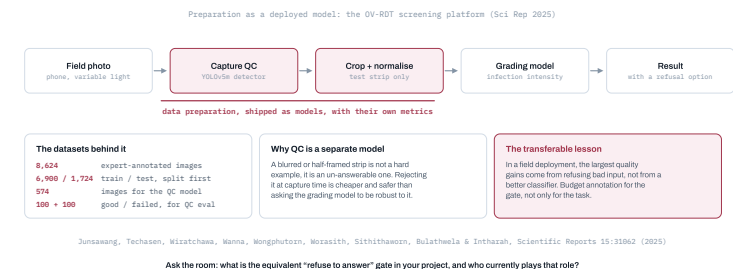
Detection: exact hashing, then perceptual hashing or MinHash/LSH for the near cases, then embedding nearest-neighbours for the semantic ones. Cheap; do it before you split.

In clinical imaging the grouping unit is the **patient** at minimum, and the **site or scanner** for any generalisation claim.

In video and CCTV the unit is the **camera and the session** — never the frame. Frames within a clip are almost perfectly correlated.

In text it is the **document or the author**, and for RAG evaluation it is the **source document**, not the chunk.

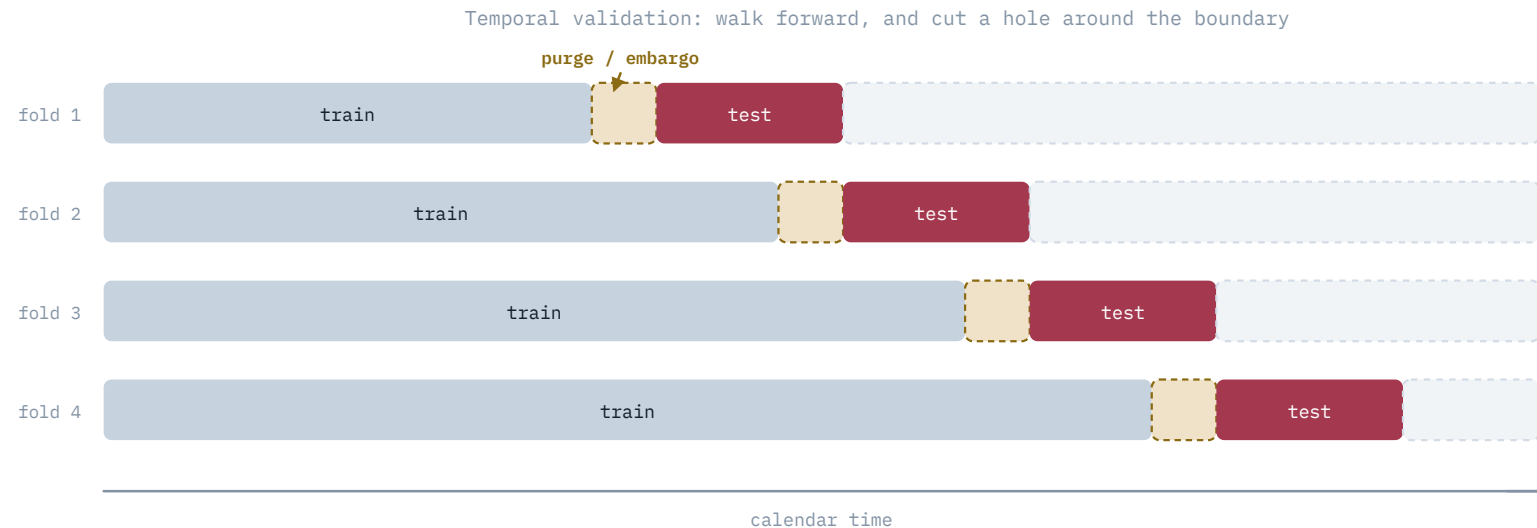
The test: if two rows would still be in the dataset had you sampled one fewer patient, camera or document, they belong to the same group.



In our own OV-RDT work the grouping unit is the specimen, and capture quality is handled by a separate model before grading — a preparation decision that carries its own dataset and its own metric.

Temporal validity: walk forward and cut a hole

For anything with a time index, the only honest evaluation trains on the past and tests on the future.



Why the gap exists

Any feature computed over a window that straddles the boundary — a 30-day rolling mean, a centred filter, a label defined by a future event — leaks the test period into training.

The two classic silent failures

A centred rolling window (uses the future by construction), and target encoding computed over the whole series. Both look like ordinary feature engineering.

The purge/embargo band exists because features and labels have width in time. López de Prado's treatment is the clearest, though the problem is not specific to finance.

Contamination, test-set reuse, and the fix

BENCHMARK CONTAMINATION

The eval set is in the training corpus

At web scale it is the default, not the exception

Detect with n-gram overlap and embedding-based matching

Decontaminate *before* training, and publish a contamination report

Revisited in Chapter 2 for pretraining corpora

ADAPTIVE OVERFITTING

A thousand papers tuned on one test set

Each paper is honest; the community is not

Recht et al.² (2019) rebuilt CIFAR-10 and ImageNet test sets and saw accuracy drop across the board

Reassuringly, the *ranking* largely survived — so relative claims aged better than absolute ones

Argues for fresh held-out sets, not just bigger ones

THE FIX

Model info sheets

A fill-in template forcing authors to state, **per leakage type**, why their setup is clean

Kapoor & Narayanan³ propose it precisely because informal reporting failed

Cheap to adopt: one page, eight rows

Make it a required appendix in every project you supervise

¹ Krizhevsky et al. *ImageNet Classification with Deep Convolutional Neural Networks*. NeurIPS, 2012.

² Recht et al. *Do ImageNet Classifiers Generalize to ImageNet?*. ICML, 2019.

³ Kapoor & Narayanan. *Leakage and the Reproducibility Crisis in Machine-Learning-Based Science*. Patterns 4(9), 2023.

Section 0.3 — references

- 1 Kapoor & Narayanan. **Leakage and the Reproducibility Crisis in Machine-Learning-Based Science**. *Patterns* 4(9), 2023 (arXiv:2207.07048). *Anchor reading for the whole module*.
- 2 Cawley & Talbot. **On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation**. *JMLR*, 2010.
- 3 Varma & Simon. **Bias in error estimation when using cross-validation for model selection**. *BMC Bioinformatics*, 2006.
- 4 Ambroise & McLachlan. **Selection bias in gene extraction on the basis of microarray gene-expression data**. *PNAS*, 2002.
- 5 Recht, Roelofs, Schmidt & Shankar. **Do ImageNet Classifiers Generalize to ImageNet?** *ICML*, 2019.
- 6 Hastie, Tibshirani & Friedman. **The Elements of Statistical Learning**, ch. 7. 2009.
- 7 López de Prado. **Advances in Financial Machine Learning**, ch. 7 (purging and embargo). 2018.
- 8 Pedregosa et al. **Scikit-learn**. *JMLR*, 2011 — for `Pipeline` and `ColumnTransformer` semantics.

Find the leakage

Bring one published applied-ML paper from your own field — ideally one you have cited, or one you were about to.

Ten minutes, alone. Fill in a model info sheet: for each of the eight leakage types, state what the paper says, or that it says nothing.

Fifteen minutes, in pairs. Swap papers. Your partner argues the case *for* the paper; you argue against. Then swap sides.

Twenty minutes, whole room. Three presentations of three minutes: name the type, quote the sentence, say whether the headline result survives.

Five minutes. Vote: how many of the papers in the room reported the grouping unit explicitly?

Ground rules: be specific and be fair. “The split is unclear” is a finding; “this paper is bad” is not. Most of these authors made an ordinary mistake in a field-wide blind spot.

TAKE HOME

Fill in the same sheet for **your own** current project before session 2. Bring the row you could not complete.

KEY TAKEAWAYS

The five sentences from today

Curation is now the experimental variable. At a fixed model, systematic data improvement is the higher-yield and far less explored lever.

A quality dimension you cannot compute is an adjective. Write the assertion, make the pipeline stop when it fails.

The grouping unit is a modelling decision. Declare it before the first experiment: patient, site, camera, session, document, author.

Fit every stateful transform inside the fold. Enforce it architecturally, not by remembering.

Report both numbers. “Random split 0.94, patient-grouped 0.79” is a stronger paper than either alone, because it shows you knew to look.

ON THE BOARD AS THEY LEAVE

The number in your abstract is a property of your split design, your label quality and your fold discipline at least as much as it is a property of your model.

NEXT

Where this goes

The next three sessions are technique catalogues. Today was the discipline that makes them mean something.

SESSION 2

Tabular data

Missingness mechanisms, encoding, the imbalance-versus-calibration argument, tabular foundation models, EHR cohorts.

Every transform today gets fitted inside a fold

SESSION 3

Text and LLM data

Thai segmentation, subword tokenisation and its cost, web-corpus filtering, deduplication, instruction and RAG data.

Deduplication is both a quality and a leakage control

SESSION 4

Image and video data

DICOM and whole-slide handling, intensity and stain normalisation, augmentation policy, CCTV and medical imaging.

Patient- and camera-level grouping, applied

SESSION 5

Labels, reduction and governance

Annotation and label noise, coresets and pruning, synthetic data, privacy, fairness and documentation.

Where the capstone brief is set

Reading before session 2: Kapoor & Narayanan, Patterns 2023.